



Subject: IT Project Management



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SOFTWARE PROJECT MANAGEMENT

Week 1 • Introduction to Software Project Management

LEARNING FOCUS

Projects • Project Manager • Constraints • Life Cycle • Stakeholders

Week 1 Introduction to Software Project Management

Step-by-step lesson structure

I. Learning Objectives

1. By the end of Week 1, students should be able to:
2. **Define** a software project and software project management.
3. **Explain** the characteristics of software projects.
4. **Identify** the major responsibilities of a software project manager.
5. **Explain** the project management life cycle.
6. **Identify** the major constraints affecting software projects.
7. **Analyze** common reasons why software projects succeed or fail.

Week 1 Introduction to Software Project Management

Step-by-step lesson structure

1. What is a Project?
2. What is a Software Project?
3. What is Software Project Management?
4. Why Software Project Management Matters
5. The Software Project Manager
6. Project Constraints
7. Project Life Cycle
8. Project Stakeholders
9. Practical Lab + Assignment

1. What is a Project?

A project is a temporary activity undertaken to create a unique product, service, or result.

A **project** is a temporary activity undertaken to create a unique product, service, or result.

Key characteristics

- Temporary
- Has a defined beginning and end
- Has specific objectives
- Produces a unique result
- Uses limited resources
- Has constraints
- Involves stakeholders

Project vs. Operations

Projects are temporary; operations are ongoing.

PROJECT • Build a new Student Information System

OPERATIONS • Run and maintain the existing system

Project success is measured against agreed objectives and deliverables

Projects end when objectives are achieved, cancelled, or no longer needed

Example

Developing a **University Student Management System** is a project because it has a defined objective, schedule, budget, team, and final deliverable.

2. What is a Software Project?

A software project is a temporary effort to develop, modify, implement, or maintain software.

A **software project** is a temporary effort to develop, modify, implement, or maintain a software system to meet defined requirements.

Examples

- Mobile Banking Application
- University LMS
- Hospital Management System
- E-Commerce Website
- Student Information System
- Hotel Management System
- Mobile Learning Application

Characteristics of Software Projects

Software projects have unique management challenges.

Characteristic	Explanation
Intangible	Software cannot be physically touched
Complex	Many technical components interact
Changing requirements	Customer requirements may change
High uncertainty	Estimates can be difficult
Team-based	Requires multiple roles
Technology dependent	Technologies can change quickly
Quality sensitive	Bugs can cause significant problems

3. What is Software Project Management?

Software Project Management (SPM) is the process of **planning, organizing, leading, monitoring, and controlling** resources and activities to achieve software project objectives within agreed **scope, time, cost, and quality** constraints

Initiate the project

Plan scope, schedule, cost, resources, risk, and quality

Execute the planned work

Monitor and control performance

Close and evaluate the project

Simple Model

Initiate → Plan → Execute → Monitor & Control → Close

Why Do We Need SPM?

Good developers alone do not guarantee a successful project.

- Controls project scope and expectations
- Improves schedule and resource management
- Identifies and manages risks
- Supports quality and customer satisfaction
- Improves communication and decision-making
- Reduces the probability of project failure

4. Why Software Projects Fail

Common causes are often management-related as well as technical.

- Unclear or changing requirements
- Poor estimation and unrealistic schedules
- Scope creep
- Insufficient resources
- Weak communication
- Poor risk management
- Inadequate quality control
- Lack of stakeholder involvement

5. Software Project Manager

A **Software Project Manager (SPM)** is responsible for coordinating people, processes, resources, schedules, risks, communication, and deliverables to achieve project objectives.

Major Responsibilities

1. Project initiation
2. Project planning
3. Scope management
4. Schedule management
5. Cost management
6. Resource management
7. Risk management
8. Quality management
9. Communication management
10. Stakeholder management
11. Change management
12. Project monitoring and control
13. Project closure

Project Manager Skills

Effective SPM requires a combination of technical and people skills.

Technical Skills + Management Skills + People Skills

Skill	Example
Technical	Understand software development
Planning	Create project schedules
Leadership	Lead the project team
Communication	Communicate with clients
Problem solving	Resolve project issues
Risk management	Identify project risks
Negotiation	Manage conflicts
Decision making	Select appropriate solutions

6. Project Constraints

Project constraints define the limits within which the project must operate.

Scope • What must be delivered?

Time • When must it be delivered?

Cost • How much can be spent?

Quality • How well must it perform?

Resources • Who and what are available?

6. Project Constraints

Project constraints define the limits within which the project must operate.

Scope – Time – Cost

These three constraints affect each other.

For example:

If the customer increases the **scope**, the project may require more **time** and **cost**.
Also introduce **Quality** as a critical outcome influenced by these constraints.

Example

A university asks:

"Add 10 new features before next month without increasing the budget."

The Project Management Triangle

Changing one constraint can affect the others.

Increase scope → may require more time or cost

Reduce time → may require more resources or cost

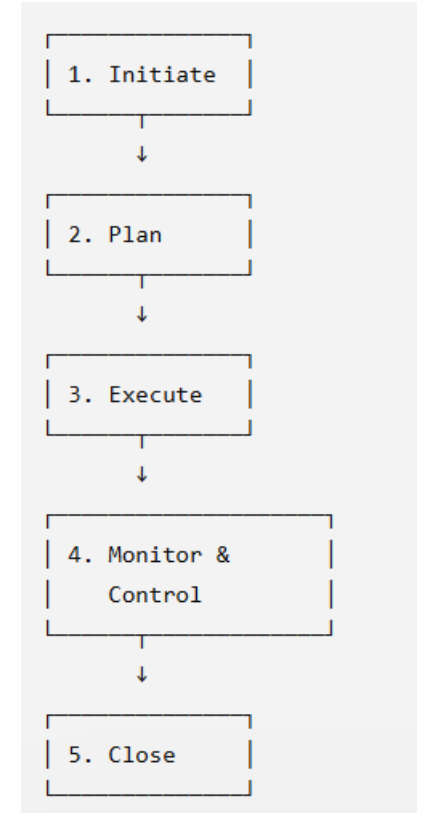
Reduce cost → may require reduced scope or longer time

Quality must be protected while balancing constraints

7. Software Project Life Cycle

A practical project-management cycle used throughout the semester.

1. Initiate → define purpose and feasibility
2. Plan → define scope, schedule, cost, resources, risks
3. Execute → perform project work
4. Monitor & Control → measure, compare, correct
5. Close → acceptance, lessons learned, handover



8. Project Stakeholders

A **stakeholder** is a person or organization that can affect, be affected by, or have an interest in the project.

Example: University LMS

Stakeholders may include:

- University management
- Students
- Lecturers
- IT department
- Project manager
- Software developers
- System administrators
- Finance department
- Software vendor

Stakeholder Example: University LMS

Ask students to classify stakeholders by influence and interest.

Management → high influence

Students → high interest

Lecturers → high interest

IT department → high influence

Vendor → technical influence

Project team → responsible for delivery

Practical Lab 1

Identify and define a software project.

Work in groups of 4–5 students

Choose a realistic software project

Write the project problem statement

Define 3–5 project objectives

Identify target users and 5–8 stakeholders

Identify major scope, time, cost, quality, and resource constraints

Practical Lab 1

Identify and define a software project.

Task

Students work in groups of **4–5 students**.
Each group selects one software project.

Step 1 — Project Title

Example:

University Student Management System

Step 2 — Problem Statement

Students identify the current problem.

Step 3 — Project Objective

Write **3–5 objectives**.

Example:

To manage student registration.

To manage student information.

To manage courses.

To generate student reports.

Practical Lab 1

Identify and define a software project.

Step 4 — Identify Users

Example:

- Students
- Lecturers
- Administrators

Step 5 — Identify Stakeholders

Students identify at least **5–8 stakeholders**.

Step 6 — Project Constraints

Identify:

- Scope
- Time
- Cost
- Quality
- Resources

Lab Deliverable

Each group produces a one-page project concept.

Project Title

Background / Problem Statement

Project Goal

3–5 Objectives

Target Users

Stakeholders

Expected Benefits

Major Constraints

Assignment 1

Software Project Idea & Problem Statement

Prepare 2–3 pages

Include title, background, problem statement, goal, objectives, users, stakeholders, benefits, and constraints

Submit as Word/PDF

Use clear and measurable objectives

Quick Quiz

Check understanding before leaving the class.

1. What makes a project temporary?
2. Give two examples of software projects.
3. Name three project constraints.
4. What are the five major project life-cycle stages?
5. Name three responsibilities of a software project manager.
6. Who are stakeholders in a university LMS project?

Week 1 Recap

The foundation for the rest of the semester.

Software projects are temporary and unique

SPM coordinates people, processes, resources, and activities

Project managers balance scope, time, cost, quality, and resources

The project life cycle guides management activities

Stakeholders must be identified and engaged

Next week: SDLC and software project methodologies

Class Discussion

Apply the concepts to a real software project

SCENARIO

A university asks your team to build a Student Management System in 4 months with a fixed budget. During development, management requests 10 additional features.

What constraints are affected?

What risks could appear?

Who are the key stakeholders?

What should the project manager do before accepting the change?

How could this situation lead to project failure?